



Southeast University

Department of Computer Science & Engineering

COMPLEX ENGINEERING PROJECT (CEP)

Statistical Investigation of Online Shopping Behavior

Prepared by,

Md. Abdullah Al Afif

Course Title: Statistical Methods and Probability

Course Code: STA281

Section – 8

Table of Contents

1. Introduction 3

2. Dataset Description 3-4

3. Statistical Analysis (Descriptive Statistics) 5-6

4. Visualization 7-12

5. Probability & Distribution Analysis 13-18

6. Hypothesis Testing 19-20

7. Correlation & Regression 21-25

8. Findings & Interpretation 26-28

9. Conclusion 28-29

10. References 30

1. Introduction

The rapid proliferation of e-commerce platforms has fundamentally transformed consumer purchasing behavior. Understanding how users interact with online shopping websites — and what drives them to complete or abandon a purchase — has become a critical area of both academic inquiry and commercial strategy. Statistical analysis of web session data provides quantitative evidence about the psychological, temporal, and behavioral factors that influence purchase intention.

This report presents a comprehensive statistical investigation of the Online Shoppers Purchasing Intention Dataset, which contains 12,330 web session records collected from a real-world e-commerce platform. The study employs descriptive statistics, probability theory, Bayesian reasoning, the Central Limit Theorem (CLT), hypothesis testing, correlation analysis, and regression modeling to extract meaningful insights about online consumer behavior.

The primary focus variables are: BounceRates, ExitRates, PageValues, ProductRelated, and VisitorType — with Revenue (binary indicator of purchase completion) as the key outcome variable.

The objectives of this project are threefold:

- Characterize the statistical distributions and central tendencies of browsing-related variables.
- Apply inferential statistical methods to test hypotheses about visitor behavior.
- Build and interpret regression models that predict purchase intention.

2. Dataset Description

The dataset employed in this study is the Online Shoppers Purchasing Intention Dataset, sourced from the Kaggle platform. It was originally constructed from real e-commerce website logs and is widely used in machine learning and statistical research for purchase prediction tasks.

The table below summarizes the Dataset used in this study:

Item	Value
Total Records	12,330
Total Variables	18
Numerical Variables	10
Categorical Variables	8
Missing Values	0
Duplicate Values	125
Total Records (Without Duplicate values)	12205
Purchase Sessions	1,908
Non-Purchase Session	10,297

Table 1: Dataset Overview

2.1 Dataset Overview

The dataset contains 12,330 rows and 18 columns. Each row represents one unique web browsing session by a user. There are no missing values in any column, which eliminates the need for imputation during preprocessing.

2.2 Variable Descriptions

The table below summarizes the six key variables used in this study:

Variable	Type	Description
BounceRates	Continuous	Fraction of sessions from that page where user left immediately (0–0.20)
ExitRates	Continuous	Fraction of page views that were the last in the session (0–0.20)
PageValues	Continuous	Average value of page visited prior to completing a transaction (Google Analytics metric)
ProductRelated	Integer	Number of product-related pages visited during session (0–705)
VisitorType	Categorical	New_Visitor (1,693), Returning_Visitor (10,431), Other (81)
Revenue	Binary	TRUE if session ended in a purchase (1,908 out of 12,330); FALSE otherwise

Table 2: Key Variables Used in This Study

2.3 Preprocessing Steps

The following preprocessing steps were applied before statistical analysis:

- Missing value analysis was performed using `isnull().sum()`, confirming that no missing values were present in any of the 18 variables.
- The boolean variable Revenue was converted to binary numeric values (0 = False, 1 = True) to facilitate correlation and regression analysis.
- Data types of all variables were verified to ensure consistency between numerical and categorical features.
- The categorical variable VisitorType was retained in its original format for grouping, probability calculations, and hypothesis testing.
- No missing-value imputation was required because the dataset was complete.
- 125 Duplicate rows were detected during duplicate row detection. After removing duplicate rows, the number of rows became 12,205. Although the dataset is designed such that each row represents a unique web session, duplicate records were identified during preprocessing. These duplicates were removed to ensure data integrity and consistency of statistical analysis.
- No outlier removal was performed. Extreme observations were preserved to maintain the integrity of the original e-commerce session data.

2.4 Class Distribution

Class	Count	Percentage
No Purchase (Revenue = False)	10,297	84.37%
Purchase (Revenue = True)	1,908	15.63%
Total	12,205	100.00%

Table 3: Revenue Class Distribution

3. Statistical Analysis

3.1 Descriptive Statistics

Descriptive statistics quantify the central tendency, dispersion, and shape of each variable's distribution. For a dataset $X = \{x_1, x_2, \dots, x_n\}$, the following formulas apply:

3.1.1 Measures of Central Tendency

$$\text{Mean: } \bar{x} = (1/n) \sum_i x_i$$

Median: middle value when data is sorted (Q2)

Mode: most frequently occurring value

3.1.2 Measures of Dispersion

$$\text{Sample Variance: } s^2 = (1/(n-1)) \sum_i (x_i - \bar{x})^2$$

$$\text{Standard Deviation: } s = \sqrt{s^2}$$

$$\text{Range} = \text{Max} - \text{Min}$$

$$\text{IQR} = \text{Q3} - \text{Q1}$$

3.1.3 Measures of Shape

$$\text{Skewness} = (n/((n-1)(n-2))) \sum_i ((x_i - \bar{x})/s)^3$$

$$\text{Kurtosis} = (n(n+1)/((n-1)(n-2)(n-3))) \sum_i ((x_i - \bar{x})/s)^4 - 3(n-1)^2/((n-2)(n-3))$$

Table 1 presents the complete descriptive statistics for the four focal quantitative variables, all computed from the 12,205 session records:

Statistic	BounceRates	ExitRates	PageValues	ProductRelated
Count	12,205	12,205	12,205	12,205
Mean	0.0204	0.0415	5.9496	32.0456
Std Dev	0.0453	0.0462	18.6537	44.5936
Min	0.0000	0.0000	0.0000	0.0000
25th Percentile	0.0000	0.0142	0.0000	8.0000
Median (50%)	0.0029	0.0250	0.0000	18.0000
75th Percentile	0.0167	0.0485	0.0000	38.0000
Max	0.2000	0.2000	361.7637	705.0000
IQR (Q3-Q1)	0.0167	0.0343	0.0000	30.0000

Skewness	3.1624	2.2346	6.3510	4.3334
Kurtosis	9.3340	4.6421	64.9980	31.0727

Table 4: Descriptive Statistics of Key Browsing Variables (n = 12,205)

3.2 Interpretation of Descriptive Statistics

BounceRates

The mean BounceRate is **0.0204 (2.04%)**, which is relatively low. However, the distribution is heavily right-skewed (**skewness = 3.1624**), with a **median of 0.0029**. This indicates that most pages have very low bounce rates, while a small fraction of pages exhibit much higher values approaching the maximum of **0.20**. The high **kurtosis (9.3340)** confirms a sharply peaked distribution with a heavy right tail. This pattern is characteristic of web engagement metrics where most sessions involve meaningful interaction.

ExitRates

The mean ExitRate is **0.0415 (4.15%)**, with a **median of 0.0250 (2.50%)** and a right skew of **2.2346**. ExitRates are consistently higher than BounceRates, which is expected because every session must end on some page. The **standard deviation (0.0462)** is slightly higher than the mean, indicating substantial relative variability. The **75th percentile of 0.0485 (4.85%)** suggests that most pages maintain moderate exit rates.

PageValues

PageValues exhibits the most extreme distributional properties. The **mean is 5.9496**, while the **median is 0.00**, indicating that more than 75% of sessions recorded zero PageValues, as confirmed by the **75th percentile of 0.00**. This occurs because PageValues are only non-zero for sessions that contribute to e-commerce revenue. The **maximum value of 361.7637**, together with the extremely high **skewness (6.3510)** and **kurtosis (64.9980)**, reflects the sparse and highly right-skewed nature of this variable. PageValues is therefore characterized by a small number of sessions generating exceptionally high values.

ProductRelated

The mean number of ProductRelated pages visited per session is **32.0456**, concealing substantial variation (**standard deviation = 44.5936**, **skewness = 4.3334**). The **median of 18 pages** is more representative of a typical session. Some users browse as many as **705 product-related pages** in a single session, reflecting highly engaged shoppers. The strong positive skewness arises from this small minority of intensive browsers, which elevates the mean above the median.

4. Visualization

All visualizations were generated using Python's matplotlib and seaborn libraries directly from the dataset. This section presents ten figures, each accompanied by an interpretation.

4.1 Distributions of Browsing Variables

Figure 1 presents histograms for each of the four focal variables. The red dashed line marks the mean and the blue dotted line marks the median. For all four variables, the mean substantially exceeds the median, confirming strong positive skewness.

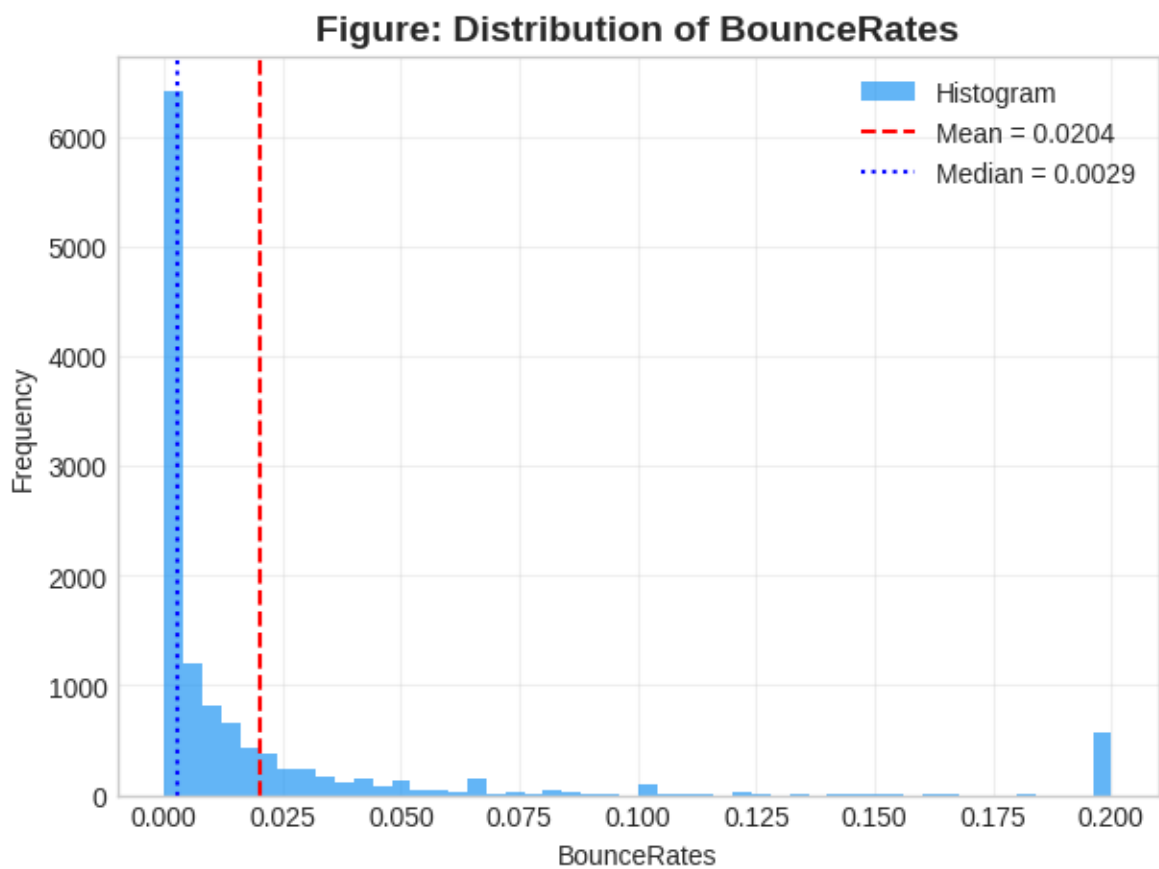


Figure 1(a): Distribution of BounceRates with Mean, and Median Lines

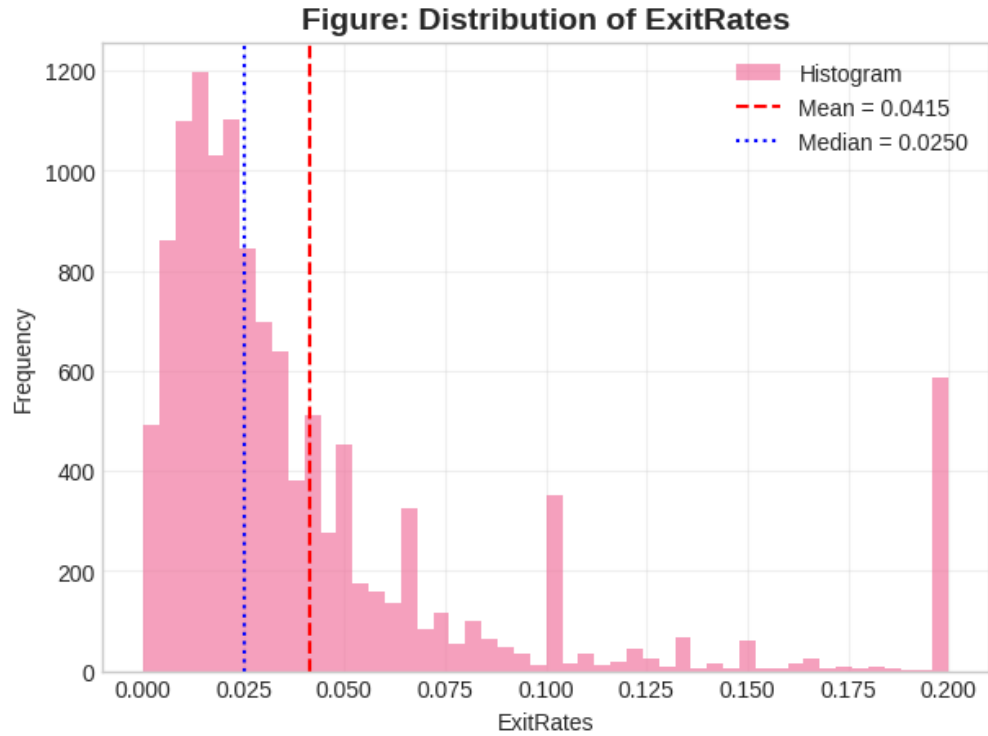


Figure 1 (b): Distribution of ExitRates with Mean, and Median Lines

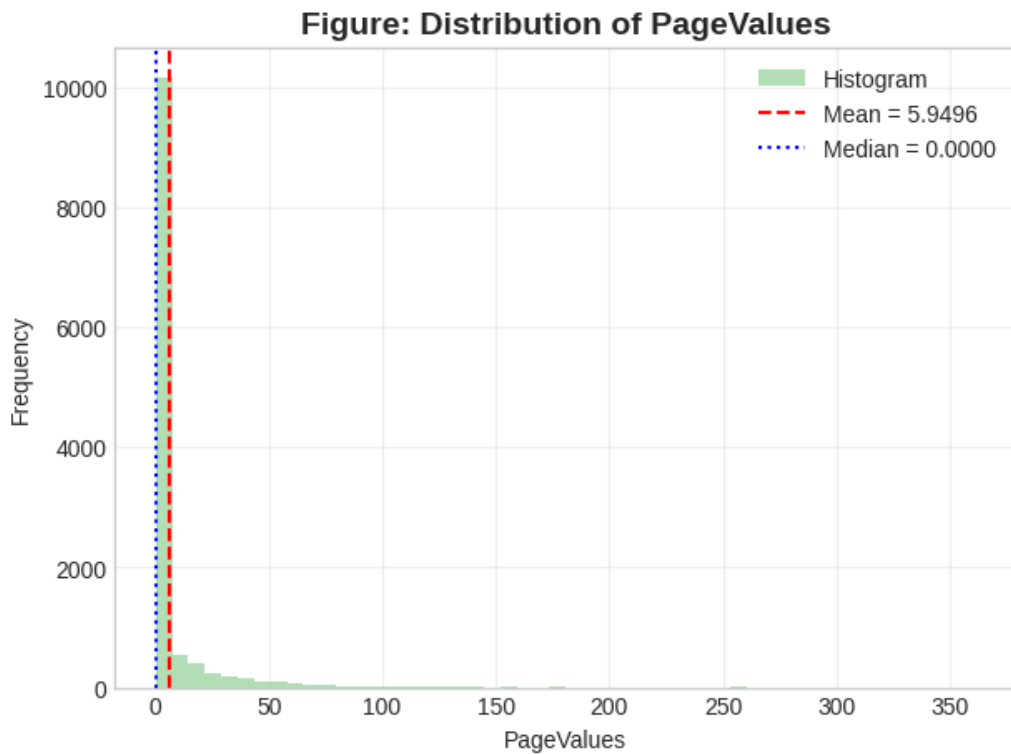


Figure 1 (c): Distribution of PageValues with Mean, and Median Lines

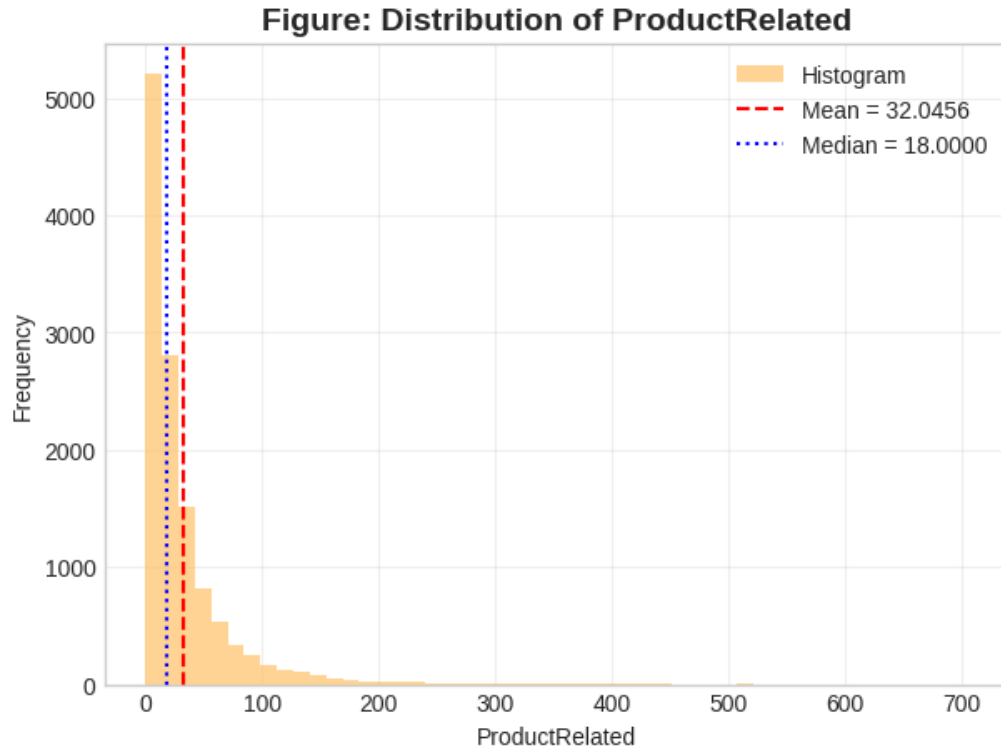


Figure 1 (d): Distribution of ProductRelated with Mean, and Median Lines

4.2 Boxplots by Purchase Outcome

Figure 2 presents boxplots for each variable grouped by whether the session resulted in a purchase (Revenue = True) or not (Revenue = False). Sessions that end in purchases show notably higher PageValues, which strongly supports PageValues as the primary predictor of purchase intention. BounceRates and ExitRates are visually lower for purchasers, consistent with more engaged browsing behavior.

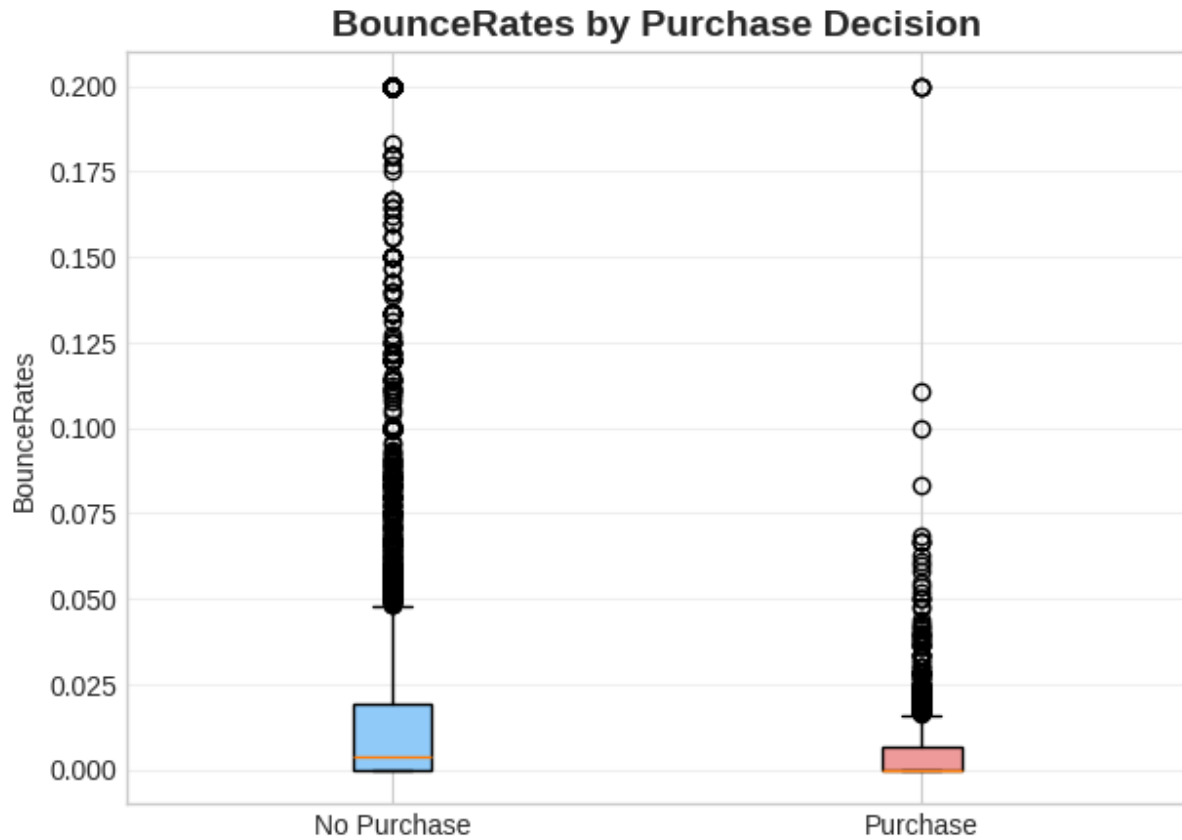


Figure 2 (a): Boxplots of BounceRate by Purchase Outcome

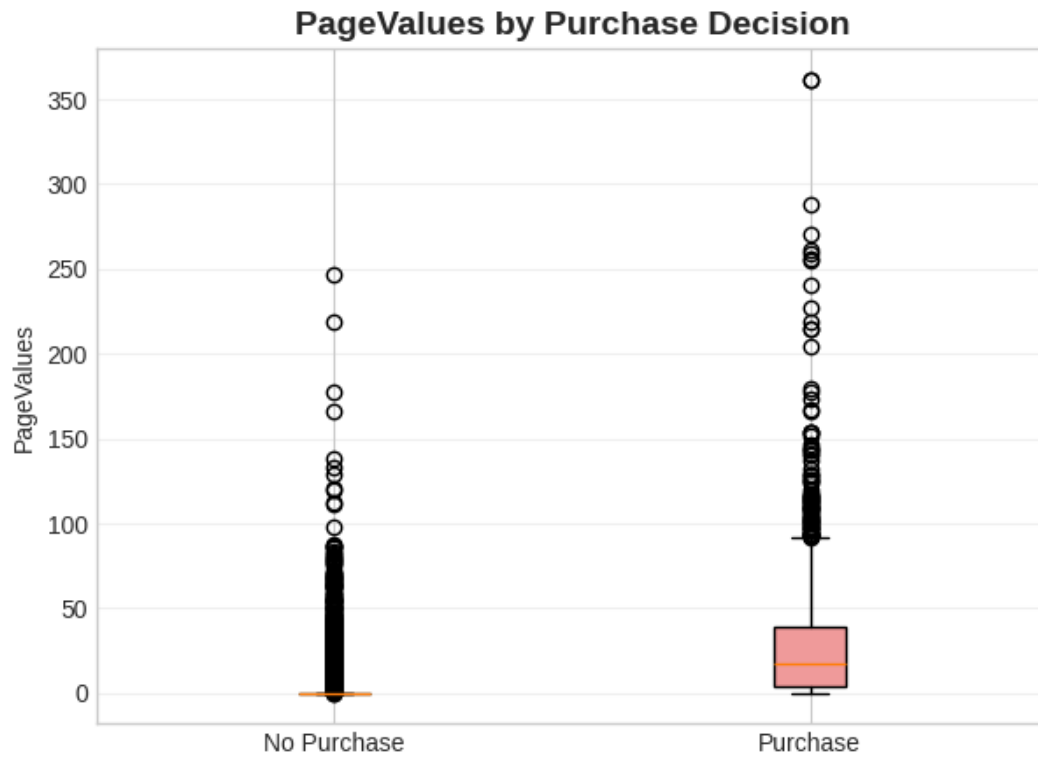


Figure 2 (b): Boxplots of PageValues by Purchase Outcome

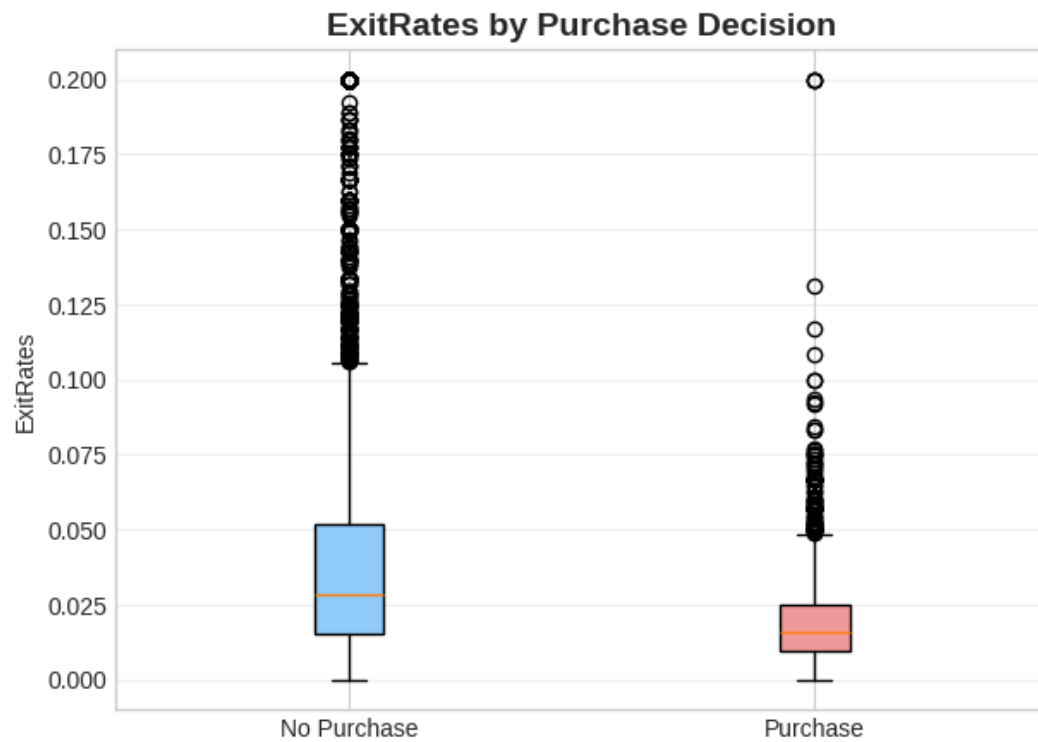


Figure 2 (c): Boxplots of ExitRate by Purchase Outcome

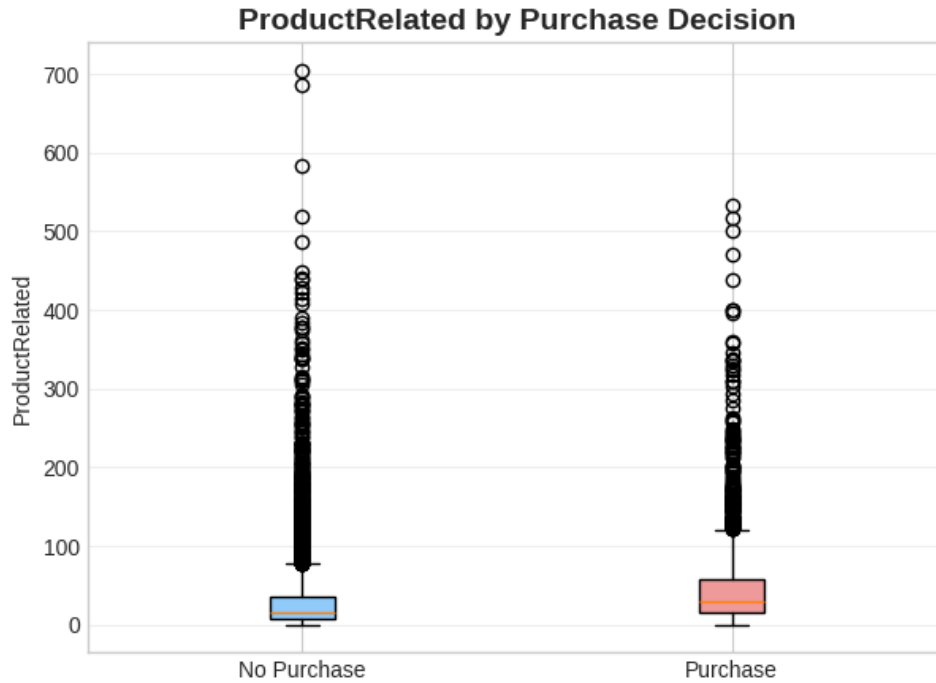


Figure 2 (d): Boxplots of ProductRelated by Purchase Outcome

4.3 Visitor Type Analysis

Figure 3a shows the distribution of visitor types. Returning visitors constitute the overwhelming majority (85.46%), while new visitors account for 13.87%. Figure 3b reveals the purchase rate by visitor type: new visitors show a substantially higher purchase rate (24.93%) compared to returning visitors (14.09%). This counter-intuitive finding may reflect those new visitors who arrive through targeted advertisements or referrals tend to have stronger purchase intent.

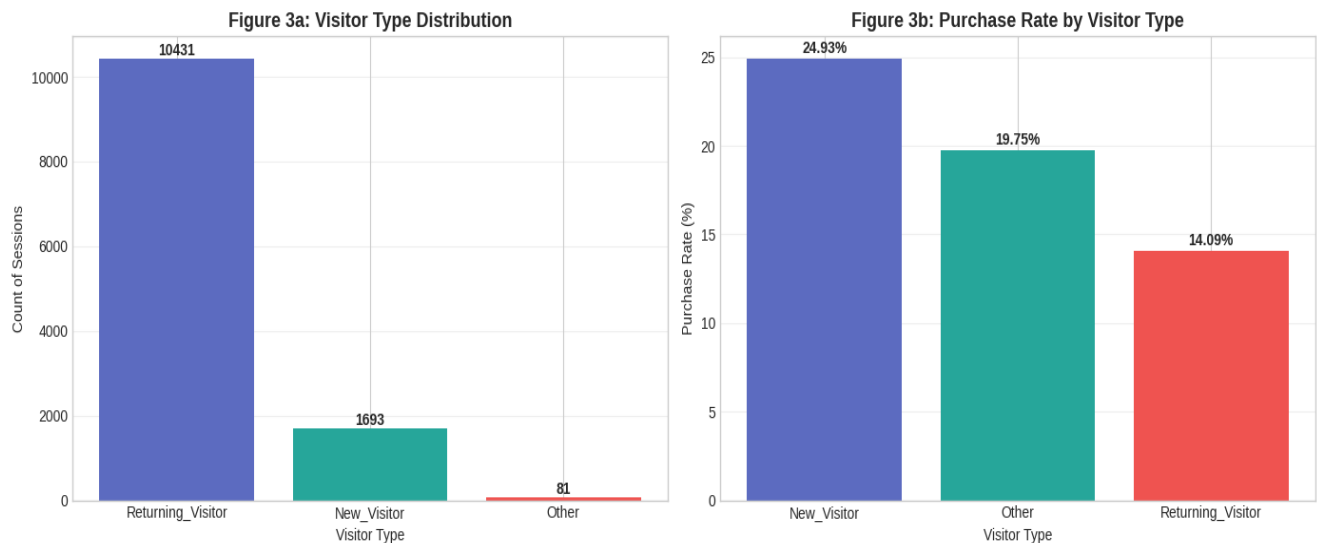


Figure 3: Visitor Type Distribution (3a) and Purchase Rate by Visitor Type (3b)

5. Probability & Distribution Analysis

5.1 Empirical Probability of Purchase

The fundamental probability of a session resulting in a purchase is computed empirically from the dataset as:

$$P(\text{Purchase}) = n_{\text{purchase}} / n_{\text{total}} = 1,908 / 12,205 = 0.156329 \text{ (15.63\%)}$$

This serves as the prior probability in subsequent Bayesian analyses. The complement:

$$P(\text{No Purchase}) = 1 - P(\text{Purchase}) = 1 - 0.156329 = 0.843671 \text{ (84.37\%)}$$

The class imbalance (approximately 5.4:1 ratio of non-purchases to purchases) is an important baseline for all predictive interpretations in this study.

5.2 Conditional Probabilities by Visitor Type

The conditional probability $P(\text{Purchase} | \text{VisitorType})$ measures the likelihood of a purchase given the type of visitor. The theorem of conditional probability is $P(A|B) = P(A \cap B) / P(B)$

Table 5 summarizes these probabilities:

Visitor Type	Count	Purchases	$P(\text{Purchase} \text{VisitorType})$	Purchase Rate
Returning_Visitor	10,431	1470	0.140926	14.09%
New_Visitor	1,693	422	0.249262	24.93%
Other	81	16	0.197531	19.75%
Overall	12,205	1908	-----	15.63%

Table 5: Conditional Purchase Probabilities by Visitor Type

New visitors have a purchase rate of 24.93%, which is approximately 1.6 times higher than the overall rate of 15.63%. Returning visitors, despite comprising 85.46% of the sample, have a below-average purchase rate of 14.09%.

5.3 Bayes' Theorem Application

Bayes' theorem allows us to reverse the conditional probability — i.e., given that a purchase occurred, what is the probability that the visitor was of a particular type? The theorem states:

$$P(A | B) = [P(B | A) \times P(A)] / P(B)$$

Applying this to estimate $P(\text{VisitorType} | \text{Purchase} [\text{Revenue} = \text{True}])$:

$$P(\text{VisitorType} | \text{Purchase}) = [P(\text{Purchase} | \text{VisitorType}) \times P(\text{VisitorType})] / P(\text{Purchase})$$

So, $P(\text{New_Visitor} | \text{Purchase}) = 0.221174 \text{ (22.12\%)}$

Similarly, $P(\text{Returning_Visitor} | \text{Purchase}) = 0.770440 \text{ (77.04\%)}$.

Interpretation: Among all sessions that result in a purchase, 22.12% came from new visitors and 77.04% from returning visitors. Despite the higher individual purchase rate for new visitors, returning visitors dominate purchase events simply because they constitute a far larger proportion of the user base.

5.4 Binomial Distribution

The number of purchases in a fixed set of n independent sessions follows a Binomial distribution. Let $X \sim \text{Binomial}(n, p)$ where $p = 0.1563$ and $n = 20$ sessions:

$$P(X = k) = C(n, k) \times p^k \times (1-p)^{(n-k)}$$

$$E[X] = np = 20 \times 0.1563 = 3.1266 \quad | \quad \text{Var}[X] = np(1-p) = 20 \times 0.1563 \times 0.8437 = 2.6378 \quad | \quad \text{Std}[X] = 1.6241$$

Selected probabilities: $P(X=0) = 0.033378$, $P(X=1) = 0.123696$, $P(X=2) = 0.217745$, $P(X=3) = 0.242084$, $P(X=4) = 0.190644$, $P(X=5) = 0.113042$. The most probable outcome is 3 purchases in 20 sessions. The complete PMF is shown in Figure 6.

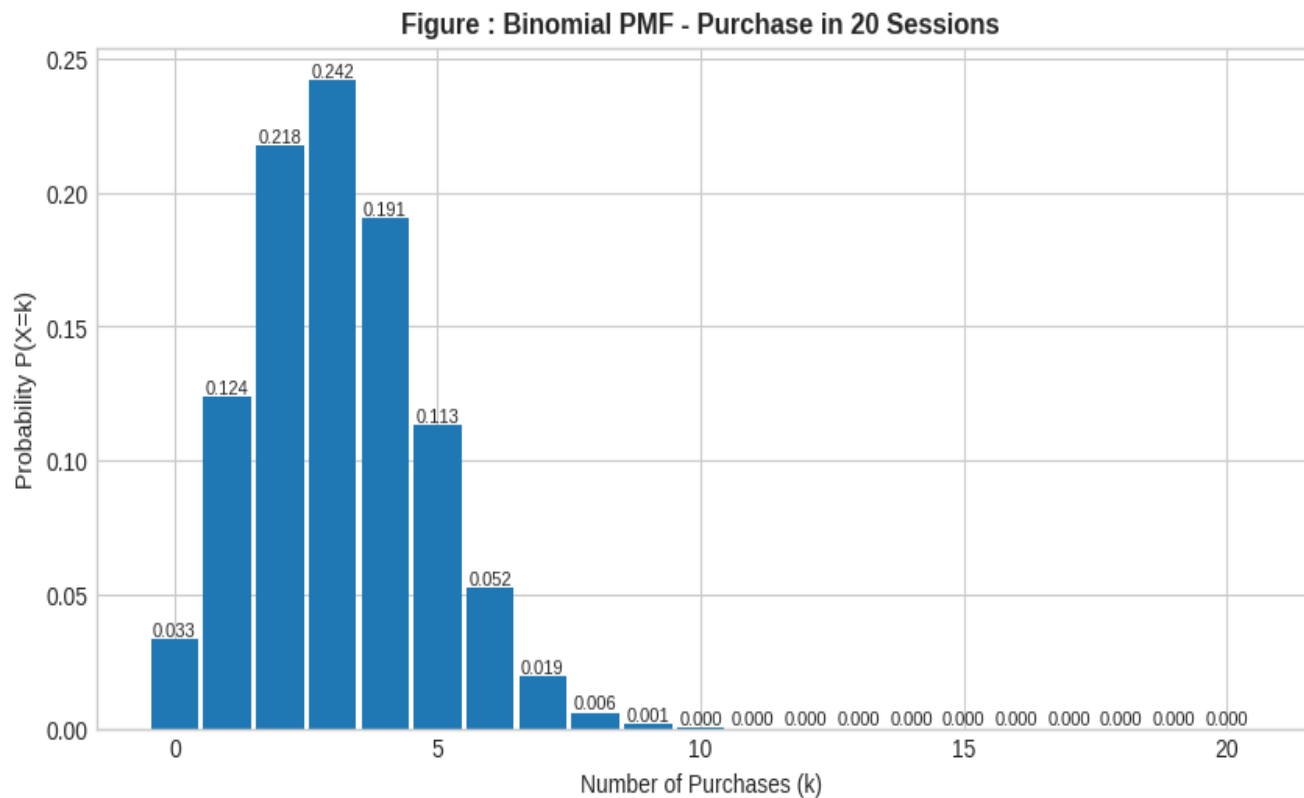


Figure 4: Binomial PMF — Probability of k Purchases in 20 Sessions ($p = 0.1563$)

5.5 Normal Distribution

The **Normal Probability Distribution** is the most important and widely used continuous probability distribution in statistics. It was significantly developed by **Karl Gauss (1777–1855)**, which is why it is also called the **Gaussian Distribution**. Normal distribution has a wide range of practical applications — human characteristics such as height, weight, and IQ scores, as well as many variables in business and industry, are normally distributed.

Probability Density Function:

$$f(x) = (1 / \sigma\sqrt{2\pi}) \times e^{-(x-\mu)^2 / 2\sigma^2}$$

Where μ is the mean, σ is the standard deviation, $e \approx 2.71828$, and $\pi \approx 3.14159$. The $f(x)$ values represent the relative frequencies (height of the curve) within which values of the random variable x occur. In notation: $X \sim N(\mu, \sigma)$.

Key Properties of Normal Distribution:

- The curve is bell-shaped and perfectly symmetrical around the mean μ
- Mean = Median = Mode — all three are equal and lie at the center
- Total area under the normal curve = 1
- $P(X \leq \mu) = P(X \geq \mu) = 0.50$ — exactly half the area lies on each side of the mean
- The two tails extend to infinity and never touch the horizontal axis
- Skewness = 0, Kurtosis = 3

5.5.1 Application to PageValues

The Normal distribution is applied to the PageValues variable using parameters computed directly from the dataset (after deduplication, $n = 12,205$):

Mean (μ) = 5.9496 Standard Deviation (σ) = 18.6537 Model: $X \sim N(5.9496, 18.6537)$

5.5.2 Empirical Rule (68 – 95 – 99.7 Rule)

The Empirical Rule states that for any Normal distribution, approximately 68%, 95%, and 99.7% of values fall within one, two, and three standard deviations of the mean respectively. Applying this to PageValues:

Range	Interval	Approximate Area
$\mu \pm 1\sigma$	$[-12.7041, 24.6032]$	$\approx 68\%$
$\mu \pm 2\sigma$	$[-31.3578, 43.2569]$	$\approx 95\%$
$\mu \pm 3\sigma$	$[-50.0114, 61.9106]$	$\approx 99.7\%$

Table :6

Note: The negative lower bounds in the intervals above reflect the theoretical nature of the Normal model. In reality, PageValues cannot be negative (minimum = 0). This indicates that the Normal distribution is an approximation for this variable, which is actually right-skewed — a limitation acknowledged in the descriptive statistics (Section 3).

5.5.3 Z-Score and Standard Normal Conversion

Every Normal distribution can be converted to the **Standard Normal Distribution** with mean 0 and standard deviation 1 using the Z-score formula:

$$Z = (x - \mu) / \sigma$$

The Z-score tells how many standard deviations a value x is above or below the mean. If x is greater than μ , Z is positive. If x is less than μ , Z is negative. If x equals μ , Z is zero.

Worked Example: $P(\text{PageValues} > 20)$

$$Z = (20 - 5.9496) / 18.6537 \quad Z = 0.7532$$

$$P(X > 20) = 1 - \Phi(0.7532) = 0.2257 = 22.57\%$$

Interpretation: Under the Normal model, approximately **22.57%** of sessions are expected to have PageValues greater than 20. Since non-zero PageValues are strongly associated with purchase intent (as confirmed in Sections 6 and 7), this indicates that roughly one in four sessions involves meaningful revenue-generating browsing behavior.

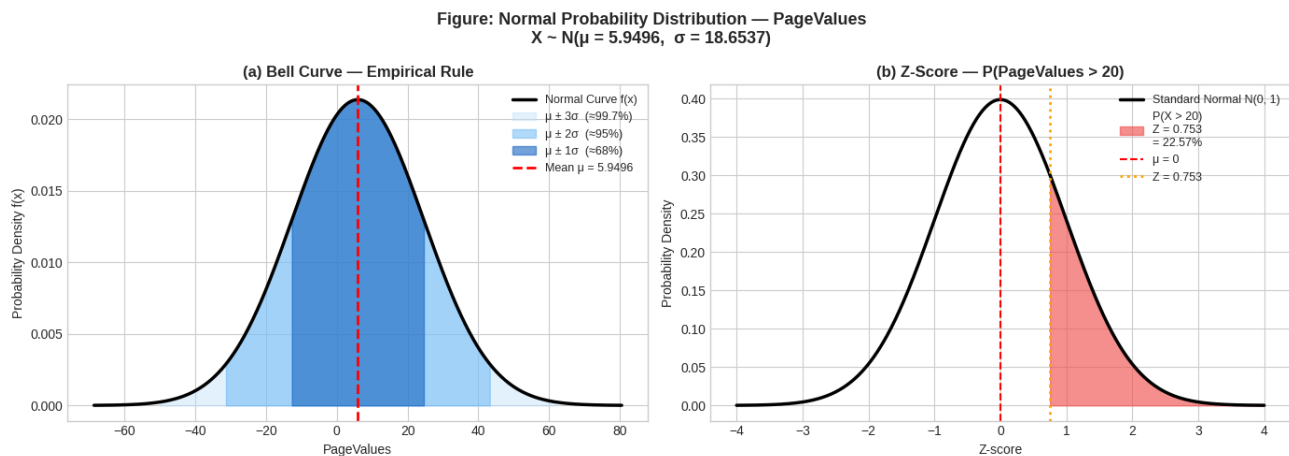


Figure 5: Normal Probability Distribution – PageValues

Interpretation of Figure: Figure (a) shows the bell-shaped Normal curve fitted to PageValues with the three Empirical Rule regions shaded — 68% (dark blue), 95% (medium blue), and 99.7% (light blue) — centered at the mean $\mu = 5.9496$. Figure (b) shows the Standard Normal curve $N(0,1)$ with the shaded red area representing $P(\text{PageValues} > 20)$, corresponding to $Z = 0.7532$, which equals 22.57% of the total area under the curve.

5.6 Central Limit Theorem (CLT) Analysis

The Central Limit Theorem states that for a population with mean μ and standard deviation σ , the distribution of sample means $\bar{x}$ computed from samples of size n approaches a Normal distribution as $n \rightarrow \infty$:

$$\bar{x} \sim N(\mu, \sigma^2/n) \rightarrow \text{Standard Error: } SE = \sigma/\sqrt{n}$$

For PageValues (population: $\mu = 5.9496$, $\sigma = 18.6537$), the theoretical SE for samples of $n = 50$:

$$SE = 18.5684 / \sqrt{50} = 2.6380$$

A Monte Carlo simulation of 5,000 samples of size 50 was conducted. The empirical mean of sample means was **5.9536** and the empirical standard deviation of sample means was **2.6177**, closely matching the theoretical values. Figure 5 demonstrates this convergence visually.

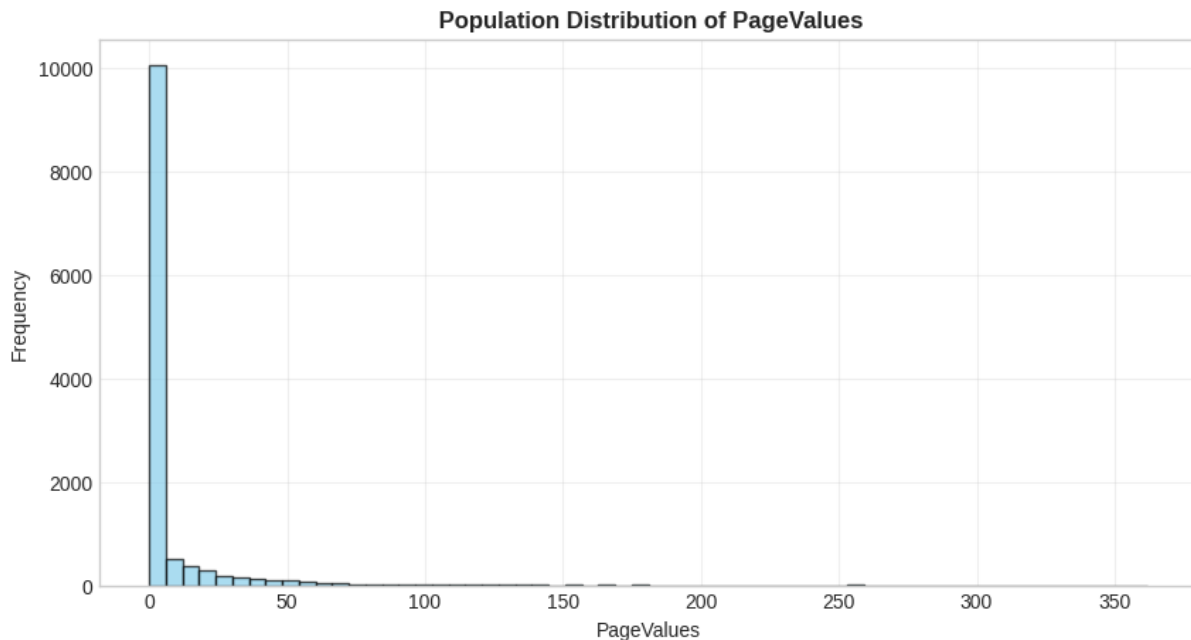


Figure 6a: Central Limit Theorem — Population Distribution

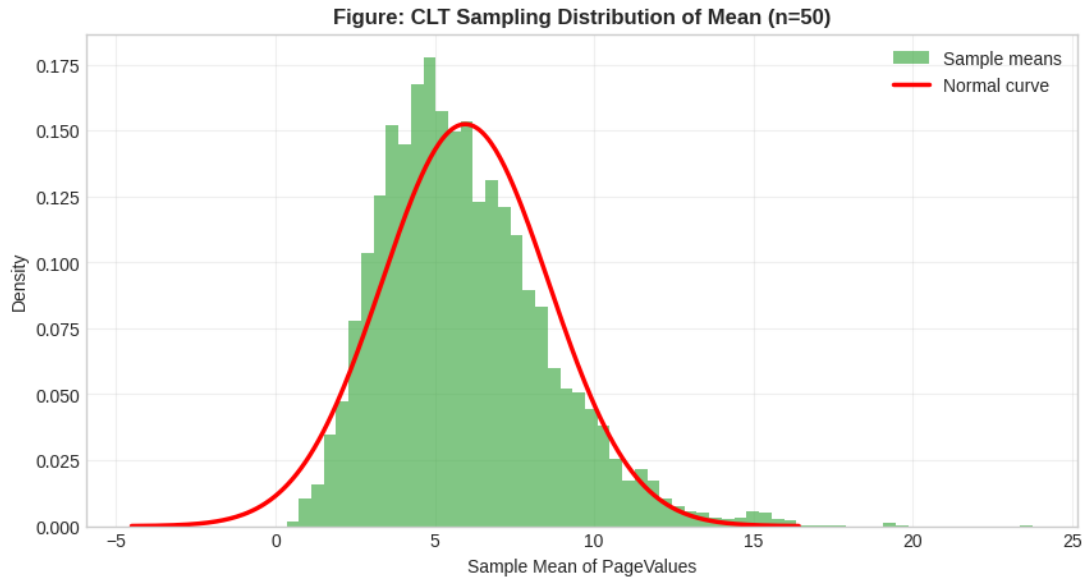


Figure 6b: Central Limit Theorem — Sampling Distribution of the Mean (n=50)

The population distribution of PageValues is extremely right-skewed (Figure 5a), yet the sampling distribution converges to a near-normal bell curve (Figure 5b). This confirms the CLT's guarantee: regardless of the population shape, the sampling distribution of the mean normalizes with sufficient sample size.

6. Hypothesis Testing

Hypothesis testing provides a formal framework for making statistical decisions. All tests use a significance level of $\alpha = 0.05$. The test statistic is compared to a critical value, and the p-value is compared to α .

6.1 Confidence Intervals for PageValues

A confidence interval (CI) provides a plausible range for the true population mean μ . Since the sample size is very large ($n = 12,205$), the Central Limit Theorem guarantees that the sampling distribution of $\bar{x}$ is approximately normal, so Z critical values are used:

$$CI = \bar{x} \pm Z_{(\alpha/2)} \times (s / \sqrt{n})$$

For PageValues: $\bar{x} = 5.9496$, $s = 18.6537$, $n = 12,205$, $SE = s/\sqrt{n} = 0.168848$

Confidence Level	Z critical	Lower Bound	Upper Bound	Width
90%	1.645	5.6718	6.2273	0.5555
95%	1.960	5.6186	6.2805	0.6619
99%	2.576	5.5146	6.3845	0.8699

Table 7: Confidence Intervals for Mean PageValues at Different Levels

The 95% CI is (5.6186, 6.2805). We are 95% confident that the true mean PageValues in the population lies within this interval. The large sample size ($n = 12,205$) produces a relatively narrow interval despite the high variability in PageValues.

6.2 Hypothesis Test 1: One-Sample Z-test — BounceRates

We test whether the dataset's mean BounceRate significantly differs from an assumed industry benchmark of 2% ($\mu_0 = 0.02$). Since $n = 12,205$ is large, the one-sample Z-test is appropriate by the Central Limit Theorem.

$$H_0: \mu_{\text{BounceRates}} = 0.02 \text{ (Two-tailed)}$$

$$H_1: \mu_{\text{BounceRates}} \neq 0.02$$

$$Z = (\bar{x} - \mu_0) / (s/\sqrt{n}) = (0.020370 - 0.02) / 0.000410 = 0.9040$$

Result: $Z = 0.9040$, $p\text{-value} = 0.3660 > 0.05$.

Decision: **FAIL TO REJECT H_0** . The mean BounceRate in this dataset (2.04%) is not statistically significantly different from the 2% industry benchmark. There is insufficient evidence to conclude that the site's bounce behavior deviates from the standard.

6.3 Hypothesis Test 2: One-Sample Z-test — PageValues

We test whether New Visitors' mean PageValues significantly differs from the overall dataset mean ($\mu_0 = 5.9496$).

$$H_0: \mu_{\text{NewVisitor}} = 5.9496 \text{ (Two-tailed)}$$

$$H_1: \mu_{\text{NewVisitor}} \neq 5.9496$$

$$Z = (\bar{x}_{\text{NV}} - \mu_0) / (s_{\text{NV}} / \sqrt{n_{\text{NV}}}) = (10.7786 - 5.9496) / 0.709599 = 6.8052$$

Result: $Z = 6.8052$, $p\text{-value} \approx 0.0000 < 0.05$.

Decision: **REJECT H_0** . New visitors browse pages with significantly higher average PageValues than the overall site average. This indicates that new visitors, likely arriving through targeted advertisements or promotions, tend to land on high-value product pages more frequently than the average session.

6.4 Hypothesis Test 3: One-Sample Z-test — ExitRates

We test whether sessions that resulted in a purchase have a mean ExitRate significantly different from the overall dataset mean ($\mu_0 = 0.0415$).

$$H_0: \mu_{\text{ExitRate(Purchase)}} = 0.0415 \text{ (Two-tailed)}$$

$$H_1: \mu_{\text{ExitRate(Purchase)}} \neq 0.0415$$

Result: $Z = -58.1330$, $p\text{-value} \approx 0.0000 < 0.05$.

Decision: **REJECT H_0** . Purchasing sessions have a significantly lower ExitRate than the overall average. Visitors who complete a purchase are far less likely to exit pages midway, confirming that low ExitRates are a strong behavioral signal of purchase intent.

Test	H0	Z-statistics	p-value	Decision ($\alpha=0.05$)
One-sample Z	$\mu_{\text{BR}} = 0.02$	$Z = 0.9040$	0.3660	FAIL TO REJECT H_0
One-sample Z	$\mu_{\text{NV(PageValues)}} = 5.9496$	$Z = 6.8052$	0.0000	REJECT H_0
One-sample Z	$\mu_{\text{buy(ExitRates)}} = 0.0415$	$Z = -58.1330$	0.0000	REJECT H_0

Table 8: Summary of Hypothesis Tests

7. Correlation & Regression

7.1 Pearson Correlation Analysis

The Pearson correlation coefficient r measures the linear relationship between two variables. It ranges from -1 (perfect negative linear relationship) to $+1$ (perfect positive linear relationship), with 0 indicating no linear association:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \times \sqrt{\sum (y_i - \bar{y})^2}}$$

Variable	BounceRates	ExitRates	PageValues	ProductRelated
BounceRates	1.0000	0.9021	-0.1158	-0.1929
ExitRates	0.9021	1.0000	-0.1736	-0.2861
PageValues	-0.1158	-0.1736	1.0000	0.0542
ProductRelated	-0.1929	-0.2861	0.0542	1.0000

Table 9: Pearson Correlation Matrix

Key findings:

- **BounceRates and ExitRates:** $r = 0.9021$ — very strong positive correlation, showing both variables capture similar user drop-off behavior.
- **ExitRates and ProductRelated:** $r = -0.2861$ — moderate negative correlation, indicating that higher product page engagement is associated with lower exit rates.
- **BounceRates and ProductRelated:** $r = -0.1929$ — weak negative correlation, suggesting slightly improved engagement with increased browsing activity.

7.2 Correlation with Revenue

The Pearson correlation with Revenue (binary: 0/1) shows:

- **PageValues ($r = 0.4919$, $p < 0.001$):** strongest predictor, showing a clear positive relationship with purchase likelihood.
- **ProductRelated ($r = 0.1560$, $p < 0.001$):** weak positive relationship, indicating browsing depth has a mild effect on purchases.
- **ExitRates ($r = -0.2043$, $p < 0.001$):** weak negative relationship, where higher exit rates reduce purchase probability.
- **BounceRates ($r = -0.1451$, $p < 0.001$):** weak negative relationship, also indicating reduced purchase likelihood with higher bounce behavior.

Overall, **PageValues stands out as the most influential variable**, while other features show weaker but statistically significant relationships with Revenue.

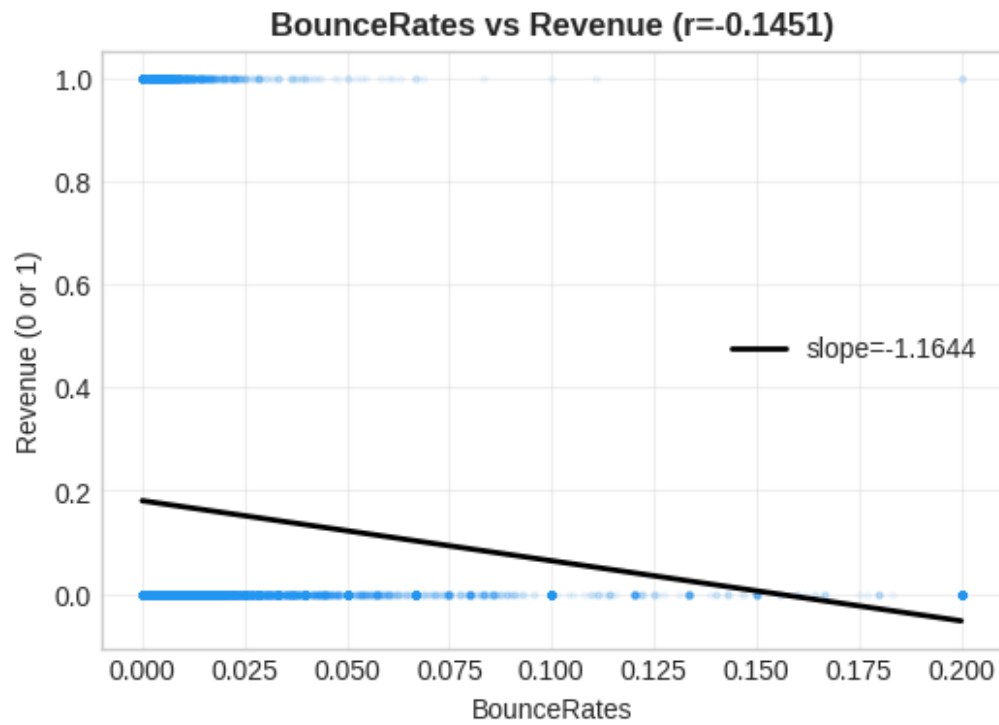


Figure 7a: Scatter Plots with Regression Lines for BounceRate vs. Revenue

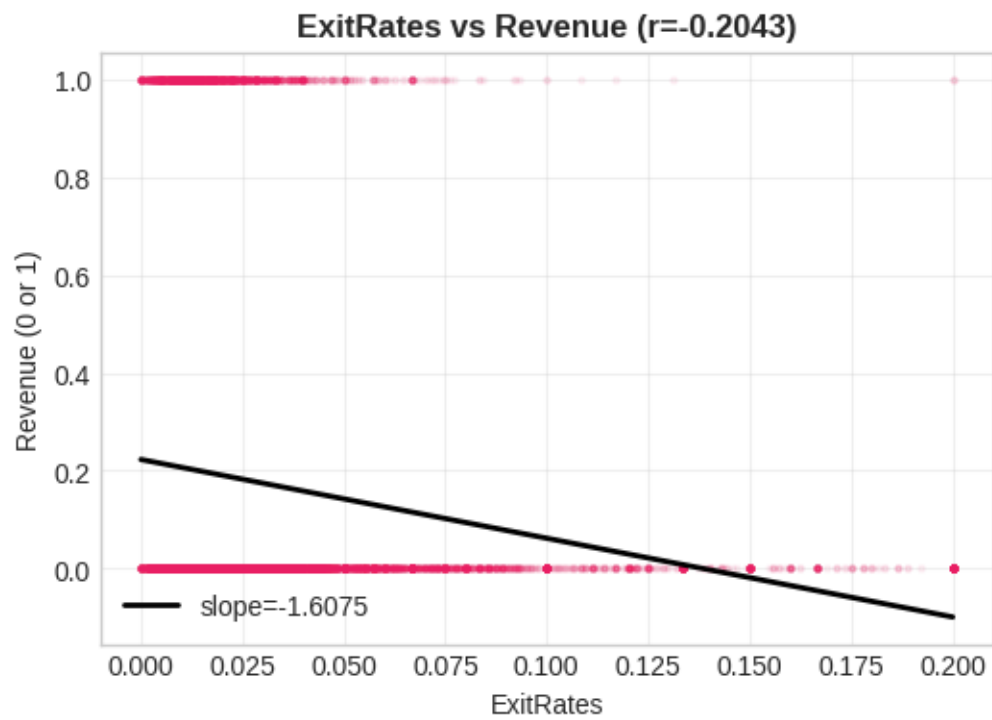


Figure 7b: Scatter Plots with Regression Lines for ExitRate vs. Revenue



Figure 7c: Scatter Plots with Regression Lines for PageValues vs. Revenue

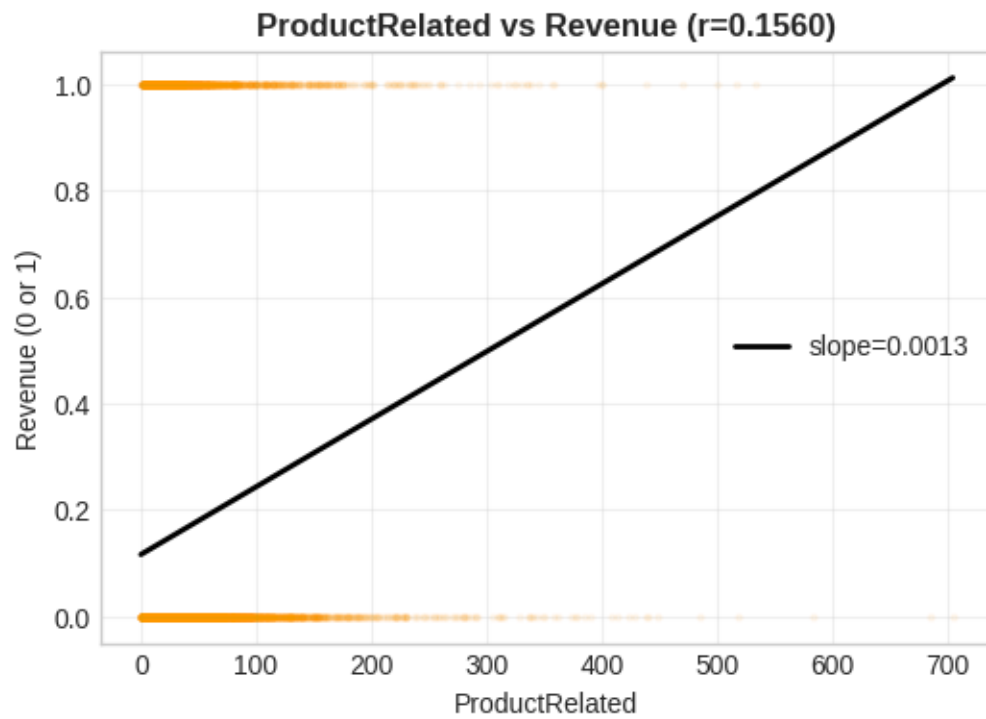


Figure 7d: Scatter Plots with Regression Lines for ProductRelated vs. Revenue

7.3 Simple Linear Regression: PageValues → Revenue

We first model the relationship between PageValues (the single strongest predictor) and the probability of purchase:

$$\text{Revenue} = \beta_0 + \beta_1 \times \text{PageValues} + \varepsilon$$

Estimated model: $\text{Revenue} = 0.0994 + 0.0096 \times \text{PageValues}$

Both coefficients are highly significant ($p < 0.001$). The $R^2 = 0.242$, meaning PageValues alone explains 24.2% of the variance in purchase outcome.

7.4 Multiple Linear Regression

We extend the model to include all four focal predictors. The regression equation is:

$$\text{Revenue} = \beta_0 + \beta_1 \cdot \text{BounceRates} + \beta_2 \cdot \text{ExitRates} + \beta_3 \cdot \text{PageValues} + \beta_4 \cdot \text{ProductRelated} + \varepsilon$$

Variable	Coefficient	Std Error	t-stat	p-value
Intercept	0.1171	0.006	20.318	0.000
BounceRates	0.5917	0.147	4.035	0.000
ExitRates	-1.2705	0.148	-8.561	0.000
PageValues	0.0091	0.000	59.078	0.000
ProductRelated	0.0008	6.67x10 ⁻⁵	12.057	0.000

Table 10: Multiple Linear Regression Coefficients

Estimated equation:

$$\text{Revenue} = 0.1171 + 0.5917 \cdot \text{BounceRates} - 1.2705 \cdot \text{ExitRates} + 0.009093 \cdot \text{PageValues} + 0.000804 \cdot \text{ProductRelated}$$

Model fit: $R^2 = 0.2674$, Adjusted $R^2 = 0.2672$, $p < 0.001$.

Coefficient Interpretations:

- Intercept (0.1171): A session with zero values for all predictors has a baseline 11.71% predicted probability of purchase.
- ExitRates (-1.2705): The strongest negative predictor. Each 1-unit increase in ExitRates (i.e., all sessions exit from that page) decreases the predicted purchase probability by 1.27 percentage points — confirming that pages driving exits strongly suppress purchase completion.
- PageValues (0.009093): Each unit increase in PageValues raises predicted purchase probability by 0.91 percentage points, which is highly significant given that PageValues ranges from 0 to 361.
- ProductRelated (0.000804): Each additional product page visited modestly increases purchase probability. The effect is small per page but accumulates across the 32.05 average pages per session.

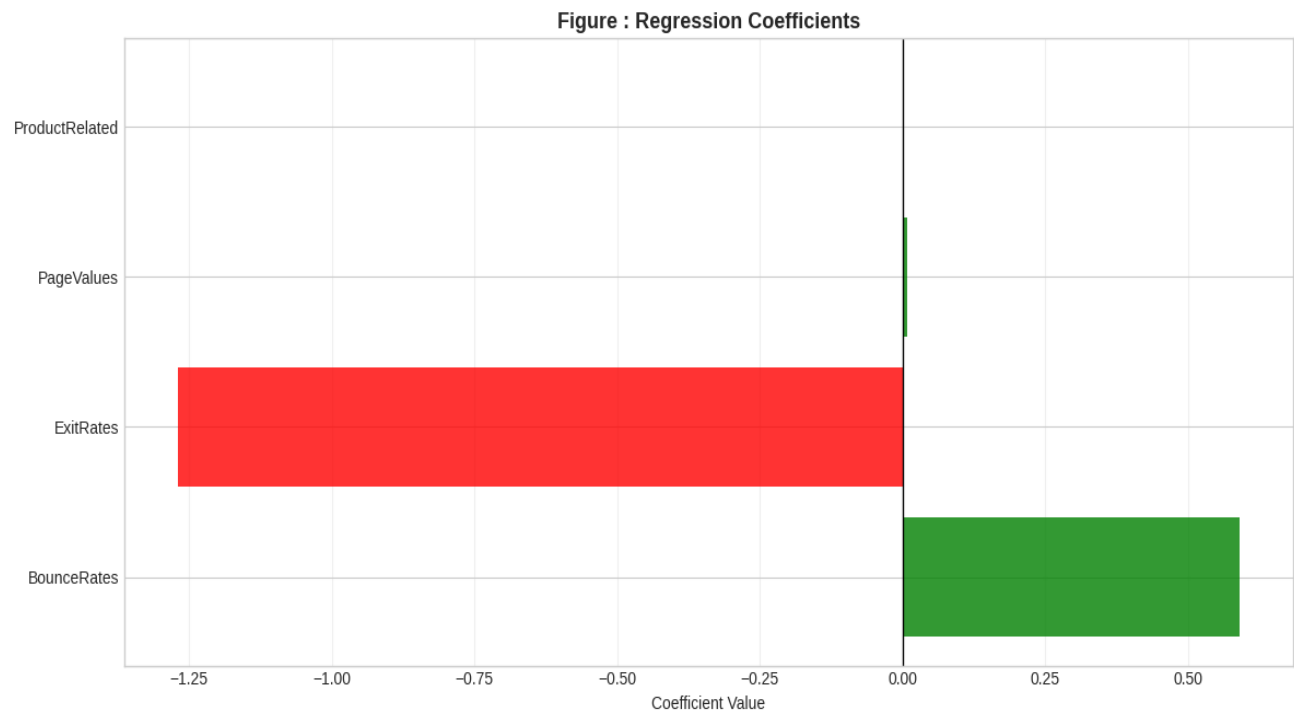


Figure 8: Regression Coefficients (Green = Positive Effect, Red = Negative Effect)

8. Findings & Interpretation

The statistical investigation of the Online Shoppers Purchasing Intention Dataset yields the following key findings, organized by analysis type:

8.1 Descriptive Findings

- All four focal variables display pronounced right-skewed distributions. BounceRates (skewness = 3.16), ExitRates (skewness = 2.23), PageValues (skewness = 6.35), and ProductRelated (skewness = 4.33) all deviate substantially from normality. For these distributions, the median is a more representative measure of central tendency than the mean.
- PageValues is the most extreme: over 75% of sessions recorded a PageValue of exactly 0.00, as confirmed by the 25th, 50th, and 75th percentiles all being zero. Only sessions that progressed toward a transaction generate non-zero PageValues, making this a highly discriminating variable.
- The average session involves browsing approximately 32 product-related pages (mean = 32.05), but the standard deviation of 44.59 and a maximum of 705 pages indicate enormous variability. A small fraction of highly engaged users heavily influences the mean, while the median of 18 pages better reflects the typical visitor.
- BounceRates and ExitRates both have means close to their respective maxima (0.0204 vs. 0.20 and 0.0415 vs. 0.20), but their high kurtosis values (9.33 and 4.64 respectively) confirm that most sessions cluster near zero with a heavy right tail of extreme values.

8.2 Probability Findings

- The empirical purchase probability is 15.63% (1,908 out of 12,205 sessions after deduplication), confirming a substantially class-imbalanced dataset with a non-purchase to purchase ratio of approximately 5.4:1. This baseline is critical for contextualizing all predictive results.
- New visitors convert at a rate of 24.93%, approximately 1.77 times higher than returning visitors (14.09%). Despite this elevated individual conversion rate, Bayesian analysis reveals that 77.04% of all purchase events originate from returning visitors, simply because they constitute 85.46% of the user base. Both acquisition and retention strategies are therefore commercially essential.
- The Binomial model ($n = 20$, $p = 0.1563$) yields an expected 3.13 purchases per 20 sessions with a standard deviation of 1.62. The most probable outcome is exactly 3 purchases ($P(X=3) = 0.2421$), consistent with the dataset's overall conversion rate.
- The Central Limit Theorem (CLT) analysis of PageValues (population: $\mu = 5.9496$, $\sigma = 18.6537$) demonstrates that sample means drawn from $n = 50$ sessions converge to a near-normal distribution with $SE = 2.638$. This convergence — despite the extreme skewness of the parent distribution — validates the theoretical basis for all Z-tests applied in this study.
- The Normal distribution was applied to PageValues with parameters $\mu = 5.9496$ and $\sigma = 18.6537$, modeled as $X \sim N(5.9496, 18.6537)$. The Empirical Rule shows that approximately 68% of sessions fall within $[-12.70, 24.60]$, 95% within $[-31.36, 43.26]$, and 99.7% within $[-50.01, 61.91]$ under the Normal model. The presence of negative lower bounds confirms that the Normal distribution is a theoretical approximation for this variable, which is non-negative and right-skewed in reality.
- Z-score analysis shows that $P(\text{PageValues} > 20) = 22.57\%$ ($Z = 0.7532$), meaning approximately one in four sessions involves meaningfully high page value browsing — a

strong behavioral indicator of purchase intent as confirmed by the regression and hypothesis testing results in Sections 6 and 7.

8.3 Hypothesis Testing Findings

- Test 1 (BounceRates vs. 2% benchmark): $Z = 0.9040$, $p = 0.3660$. Fail to reject H_0 . The site's mean BounceRate of 2.04% is not statistically distinguishable from the industry benchmark of 2%, indicating that overall bounce behavior is within expected norms.
- Test 2 (New Visitor PageValues vs. overall mean): $Z = 6.8052$, $p \approx 0.0000$. Reject H_0 . New visitors engage with pages of significantly higher average value ($\mu = 10.78$) compared to the overall site mean (5.95), suggesting targeted acquisition channels direct new users toward high-value product pages.
- Test 3 (Purchasing sessions' ExitRates vs. overall mean): $Z = -58.1330$, $p \approx 0.0000$. Reject H_0 . Sessions that result in a purchase have dramatically lower ExitRates than the dataset mean (0.0415), confirming that low ExitRates are a reliable behavioral marker of purchase intent.
- The 95% confidence interval for mean PageValues is (5.6186, 6.2805), with a width of 0.6619. The relatively narrow interval, despite high variance in PageValues ($s = 18.65$), is a direct result of the large sample size ($n = 12,205$) reducing the standard error to 0.1688.

8.4 Correlation & Regression Findings

- Pearson correlation analysis reveals a very strong positive relationship between BounceRates and ExitRates ($r = 0.9021$), confirming that both metrics capture similar aspects of user disengagement. Their collinearity has implications for regression model interpretation.
- PageValues has the strongest correlation with Revenue ($r = 0.4919$, $p < 0.001$), far exceeding all other predictors. This single variable accounts for $R^2 = 24.2\%$ of variance in purchase outcome in simple linear regression, the highest of any individual predictor.
- In multiple linear regression, ExitRates emerges as the most negative predictor ($\beta = -1.2705$), underscoring that pages driving visitor exits actively suppress conversions. BounceRates, in contrast, shows a positive coefficient ($\beta = +0.5917$) in the presence of ExitRates — likely reflecting the suppression effect caused by their high intercorrelation ($r = 0.9021$).
- The four-variable model achieves $R^2 = 0.2674$ and Adjusted $R^2 = 0.2672$. While statistically robust (F-test $p < 0.001$), approximately 73% of the variance in Revenue remains unexplained. Unmeasured factors — pricing, promotions, device type, browsing time, and brand familiarity — likely account for this residual variance.

8.5 Business Implications

The statistical findings carry the following actionable implications for e-commerce platform management:

- PageValues maximization: Since PageValues is the single strongest predictor of purchase ($r = 0.4919$), e-commerce managers should strategically feature high-margin, high-value products on landing and entry pages. Promotions that drive traffic to these pages are expected to yield measurable conversion improvements.
- ExitRate remediation: The large negative regression coefficient for ExitRates ($\beta = -1.2705$) means that even modest reductions in ExitRate on key product pages translate to meaningful predicted gains

in purchase probability. Improvements in page load speed, mobile responsiveness, clearer calls-to-action, and checkout flow simplification are recommended.

- Visitor segmentation: New visitors convert at nearly twice the rate of returning visitors. Platforms should design differentiated landing experiences: conversion-focused journeys for new visitors (who are transactionally ready) and loyalty and re-engagement strategies for returning visitors (who drive purchase volume).
- Bounce rate monitoring: Although the overall BounceRate is within industry norms, the high skewness reveals that a small subset of pages suffers from extreme bounce rates. Targeted audits of these outlier pages — for relevance, load time, and content quality — can yield disproportionate engagement improvements.

9. Conclusion

This study has conducted a rigorous, end-to-end statistical investigation of online shopping behavior using the Online Shoppers Purchasing Intention Dataset, comprising 12,205 unique web session records after deduplication. The analysis progressed systematically through data description, probability modeling (**including Binomial and Normal distributions, Bayes' Theorem, and Central Limit Theorem**), hypothesis testing, and predictive regression, yielding a coherent and internally consistent body of findings.

The central conclusion of this investigation is that PageValues is the most statistically and practically significant predictor of purchase intention. With a Pearson correlation of $r = 0.4919$ with Revenue and a regression coefficient that remains highly significant ($p < 0.001$) in the presence of three additional predictors, PageValues serves as a direct proxy for purchase-intent browsing behavior. This metric — which captures the Google Analytics-measured value of pages visited prior to a transaction — uniquely distinguishes sessions that culminate in a purchase from those that do not.

ExitRates emerged as the most consequential negative driver of purchase completion. Hypothesis Test 3 produced a Z-statistic of -58.13 ($p \approx 0.0000$), confirming an extremely significant difference in ExitRates between purchasing and non-purchasing sessions. The regression coefficient of -1.2705 for ExitRates indicates that pages prompting users to exit actively suppress conversion — making ExitRate reduction a high-priority operational target for any e-commerce platform.

The probability analysis illuminated an important tension in visitor behavior. While new visitors exhibit a substantially higher individual purchase rate (24.93% vs. 14.09% for returning visitors), Bayes' Theorem reveals that 77.04% of all purchase events are attributable to returning visitors by virtue of their numerical dominance (85.46% of sessions). This finding underscores the dual imperative of targeted new visitor acquisition and sustained returning visitor engagement.

The Central Limit Theorem demonstration validated the theoretical underpinning for all inferential methods applied in this study. Despite the extreme right-skewness of PageValues ($\gamma_1 = 6.35$), sample means of size $n = 50$ converge to a near-normal distribution with empirical statistics matching theoretical predictions (empirical SE = 2.618 vs. theoretical SE = 2.638), justifying the use of Z-tests throughout.

The multiple regression model (Adjusted $R^2 = 0.2672$) captures a moderate but statistically robust portion of variance in purchase outcome. The unexplained 73% of variance points to the inherent complexity of consumer

purchasing decisions, which are influenced by unobserved factors including product pricing, promotional incentives, device type, session timing, and individual brand perceptions. These limitations suggest several promising avenues for future research:

- Logistic regression or machine learning classifiers (Random Forest, Gradient Boosting) for improved binary outcome prediction.
- Temporal analysis incorporating Month, Weekend, and SpecialDay variables to characterize purchase seasonality.
- Cluster analysis of visitor segments based on browsing behavior profiles.
- Survival analysis of session length and drop-off timing to model abandonment dynamics.
- Incorporation of price, discount, and review data to address the primary sources of unexplained variance.

In summary, this project demonstrates that statistical analysis of e-commerce session data can yield actionable, quantitatively grounded insights. The methods applied — from descriptive statistics through Bayesian probability, **Binomial and Normal probability distributions**, CLT-based inference, hypothesis testing, and regression modeling — provide a replicable analytical framework applicable to any online retail dataset. The findings affirm that enhancing page value and reducing exit behavior are the two highest-leverage interventions for improving online purchase conversion rates.

10. References

- [1] Kaggle. (2020). Online Shoppers Purchasing Intention Dataset. Retrieved from <https://www.kaggle.com/code/vigneshprakash/online-shoppers-purchasing-intention-pca-smote>
- [2] Montgomery, D. C., & Runger, G. C. (2014). Applied Statistics and Probability for Engineers (6th ed.). John Wiley & Sons.
- [3] Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. (2012). Probability & Statistics for Engineers and Scientists (9th ed.). Pearson.
- [4] McKinney, W. (2017). Python for Data Analysis (2nd ed.). O'Reilly Media.
- [5] Pandas Development Team. (2023). pandas: Powerful Python data analysis toolkit. Retrieved from <https://pandas.pydata.org>
- [6] Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and Statistical Modeling with Python. Proceedings of the 9th Python in Science Conference.
- [7] Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), 90-95.
- [8] Waskom, M. L. (2021). seaborn: Statistical data visualization. Journal of Open Source Software, 6(60), 3021. <https://doi.org/10.21105/joss.03021>
- [9] Virtanen, P., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. Nature Methods, 17(3), 261-272. <https://doi.org/10.1038/s41592-019-0686-2>
- [10] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.